

Hypothesis Testing: Choosing the Cutoff

In class we used the simulation method: reject the hypothesis if a result as extreme as yours shows up in **about 5% or less** of the simulated samples. **If your teacher uses a different cutoff, go with your teacher's.**

What the cutoff is called

In statistics, that 5% cutoff is called the **significance level**, written $\alpha = 0.05$ (α is the Greek letter “alpha”).

The claim you are testing (e.g. “the coin is fair”) is called the **null hypothesis**.

How the decision changes with the cutoff

Teacher's cutoff	Reject the null hypothesis if the result shows up in...	What it means
1% ($\alpha = 0.01$)	1% or less of the simulated samples	Stricter: needs stronger evidence to reject
5% ($\alpha = 0.05$)	5% or less of the simulated samples	The usual standard
10% ($\alpha = 0.10$)	10% or less of the simulated samples	More lenient: easier to reject

The core decision rule (the same for every cutoff)

If $P(\text{simulated outcome} \geq \text{observed outcome}) \leq \text{teacher's cutoff}$
→ **Reject the null hypothesis.**

Otherwise → **Do not reject** it. The data doesn't give enough evidence against the claim, but that does *not* prove the claim is true.

Quick check with our coin examples

- **36 heads in 50 flips:** 0 of 200 fair-coin samples were that high, which is **0%**. That is below every cutoff (1%, 5% or 10%), so **reject** in all three cases.
- **29 heads in 50 flips:** 39 of 200 samples were that high, which is **19.5%**. That is above every cutoff, so **do not reject** in all three cases.